

Xiao Mao(毛啸)

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Education

- **Stanford University** *2022 to Present*
Ph.D.
 - Advisor: Aviad Rubinstein
 - Research Area: Theoretical Computer Science
- **Massachusetts Institute of Technology** *2021 to 2022*
M.Eng.
 - Thesis Supervisor: Virginia Vassilevska Williams
- **Massachusetts Institute of Technology** *2017 to 2021*
B.S. in Computer Science and Engineering and in Mathematics

Teaching Experience

- **Design and Analysis of Algorithms** *Fall 2025*
Course Assistant (Instructor: Mary Wootters)
- **Design and Analysis of Algorithms** *Fall 2024*
Course Assistant (Instructor: Aviad Rubinstein)
- **Design and Analysis of Algorithms (MIT 6.046J)** *Spring 2022*
Teaching Assistant
(Instructors: Srinivas Devadas, Vinod Vaikuntanathan, Virginia Vassilevska Williams)

Publications

All authors are listed in lexicographic order.

- [1] Xiao Mao and Aviad Rubinstein. Approximation schemes for Edit Distance and LCS in quasi-strongly subquadratic time. In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing*, STOC '26, New York, NY, USA, 2026. Association for Computing Machinery
- [2] Bartłomiej Dudek, Nick Fischer, Geri Gokaj, Ce Jin, Marvin Künnemann, Xiao Mao, and Mirza Redžić. Classifying Identities: Subcubic Distributivity Checking and Hardness from Arithmetic Progression Detection. In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing*, STOC '26, New York, NY, USA, 2026. Association for Computing Machinery

- [3] Ran Duan, Jiayi Mao, Xiao Mao, Xinkai Shu, and Longhui Yin. Breaking the Sorting Barrier for Directed Single-Source Shortest Paths. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing*, STOC '25, page 36–44, New York, NY, USA, 2025. Association for Computing Machinery. doi:10.1145/3717823.3718179
(STOC 2025 Best Paper) (Invited to the Journal of the ACM)
- [4] Xiao Mao. $(1 - \epsilon)$ -Approximation of Knapsack in Nearly Quadratic Time. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, STOC 2024, page 295–306, New York, NY, USA, 2024. Association for Computing Machinery. doi:10.1145/3618260.3649677
- [5] Xiao Mao. Fully Dynamic All-Pairs Shortest Paths: Likely Optimal Worst-Case Update Time. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, STOC 2024, page 1141–1152, New York, NY, USA, 2024. Association for Computing Machinery. doi:10.1145/3618260.3649695
- [6] Mingyang Deng, Ce Jin, and Xiao Mao. *Approximating Knapsack and Partition via Dense Subset Sums*, pages 2961–2979. URL: <https://epubs.siam.org/doi/abs/10.1137/1.9781611977554.ch113>, arXiv:<https://epubs.siam.org/doi/pdf/10.1137/1.9781611977554.ch113>, doi:10.1137/1.9781611977554.ch113
- [7] Mingyang Deng, Xiao Mao, and Ziqian Zhong. *On Problems Related to Unbounded SubsetSum: A Unified Combinatorial Approach*, pages 2980–2990. URL: <https://epubs.siam.org/doi/abs/10.1137/1.9781611977554.ch114>, arXiv:<https://epubs.siam.org/doi/pdf/10.1137/1.9781611977554.ch114>, doi:10.1137/1.9781611977554.ch114
- [8] Xiao Mao. Breaking the Cubic Barrier for (Unweighted) Tree Edit Distance. *SIAM Journal on Computing*, 0(0):FOCS21–195–FOCS21–223, 0. arXiv:<https://doi.org/10.1137/22M1480719>, doi:10.1137/22M1480719
SIAM Journal Version
- [9] Xiao Mao. Breaking the Cubic Barrier for (Unweighted) Tree Edit Distance . In *2021 IEEE 62nd Annual Symposium on Foundations of Computer Science (FOCS)*, pages 792–803, Los Alamitos, CA, USA, February 2022. IEEE Computer Society. URL: <https://doi.ieeecomputersociety.org/10.1109/FOCS52979.2021.00082>, doi:10.1109/FOCS52979.2021.00082
(Machtey Award for Best Student Paper, single author, sole winner) (Published in the SICOMP Special Issue for FOCS 2021)

Selected Awards and Scholarships

- **STOC 2025** 2025
Best Paper
- **FOCS 2021** 2021
Best Student Paper (Machtey Award, single author, sole winner)
- **45th ICPC World Finals** November 2022
Gold medal, 1st place
- **International Olympiad in Informatics** July to August 2017
Silver medal, 33th place
- **National Olympiad in Informatics, China** July 2016
Gold medal, 1st place

Talks

- **Breaking the sorting barrier for directed single-source shortest paths**
 - Stanford Theory Lunch July 2025
 - Fine-grained Complexity Workshop at EnCORE Institute, UCSD May 2025
 - UCB Theory Lunch May 2025

- **Fully Dynamic All-Pairs Shortest Paths: Likely Optimal Worst-Case Update Time**
– STOC 2024 *June 2024*
- **$(1 - \epsilon)$ -Approximation of Knapsack in Nearly Quadratic Time**
– STOC 2024 *June 2024*
– Stanford Theory Lunch *June 2023*
- **Approximating Knapsack and Partition via Dense Subset Sums**
– SODA 2023 *Jan 2023*
- **Breaking the Cubic Barrier for (Unweighted) Tree Edit Distance**
– FOCS 2021 *Feb 2022*
– Yao Class student seminar *Sep 2021*
– Theory seminar at the University of Washington *Mar 2022*

Research and Work Experience

- **Pika (Mellis, Inc.), Palo Alto, CA** *Summer 2025*
Intern
– AI research intern.
- **Stanford University** *Sep. 2022 to present*
Ph.D. currently advised by Prof. Aviad Rubinstein
– Focus on algorithms and complexity.
- **Massachusetts Institute of Technology** *Sep. 2021 to Sep. 2022*
M.Eng. with thesis supervised by Prof. Virginia Vassilevska Williams
– Focus on algorithms and complexity.
- **Massachusetts Institute of Technology** *Feb. 2020 to Dec. 2020*
UROP advised by Professor Michael Sipser
– Focus on algorithms and complexity.
- **Microsoft Corporation, Bellevue, WA** *Summer 2019*
Intern
– Software engineer intern.
- **Pony.ai, Inc., Fremont, CA** *Summer 2018*
Intern
– Software engineer intern.

Service

- Conference Reviewing: ITCS 2022, SWAT 2022, MFCS 2022, SODA 2024, STOC 2024, FOCS 2024, SODA 2025, SODA 2025, STOC 2025, SODA 2026, STOC 2026